

Sequential Logic Circuit

output of
sequential
circuit.

$$y = (x_1, x_2, x_3, \dots, x_n, y_{n-1}, y_{n-2}, \dots, y_{n-m})$$

Present Input
previous / past state

Asynchronous Vs synchronous

Flip Flop

(known as Latch)

Asynchronous



RS Latch

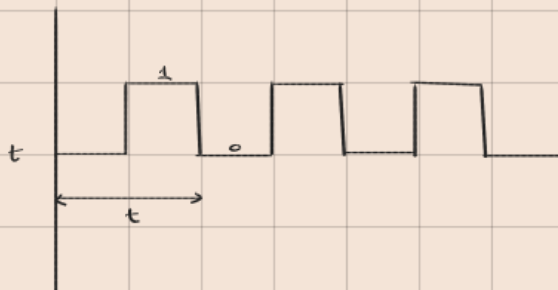
Synchronous (known as F/F)



Clock RS F/F

JK Flip Flop

Sync → Periodicity
↳ clock / enable

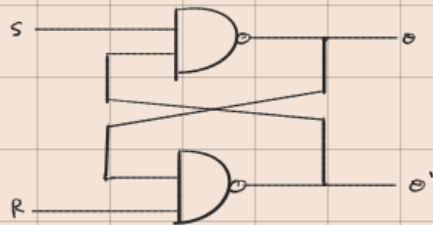


SR Latches

NAND $\rightarrow$ any i/p "0" output is "1"

NOR $\rightarrow$ any i/p "1" output is "0"

SR Latch with NAND Gate

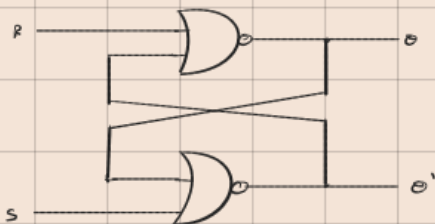


S	R	Q (new)	Q
0	0	X	X
0	1	1	0
1	0	0	1
1	1	No change	(hold state)

X = invalid / indetermined.

SR Latch with NOR Gate

SR latch with NOR Gate



S	R	Q (new)	Q
0	0	No change	(hold)
0	1	0	1
1	0	1	0
1	1	X	X

